EMPLOYER

WEST AFRICAN ENERGY SA

SENEGAL NATIONAL ELECTRICITY AGENCY

EPC CONTRACTOR

CAP DES BICHES COMBINED CYCLE POWER STATION PROJECT
(300 MW)

REVISION HISTORY

REV.	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
F	28.12.2021	FOR REVIEW	CS	TLM	SSR
E	16.09.2021	FOR REVIEW	CS	TLM	SSR
D	19.08.2021	FOR REVIEW	CS	TLM	SSR
C	21.06.2021	FOR REVIEW	CS	TLM	SSR
B	19.03.2021	FOR REVIEW	CS	TLM	SSR
A	18.01.2021	FOR REVIEW	CS	TLM	SSR

DOCUMENT TITLE/SUBJECT:

COLOR CODE, PAINTING AND COATING SPECIFICATION

PROJECT DOCUMENT NO:

Project	*Unit*	*System*	*Disc*	*Doc.*	*Type*	*Orig.*	*Serial No*
SNC	36	VDJ	MI			CEI	001

VENDOR DOCUMENT NO: *** if applicable*

Detailed Information

Rev.	F
Size	A-4
Lang.	Eng.
Total Sheet	26

1.0 COLOR CODE, PAINTING AND COATING

1.1 General

This specification covers the general requirements related to the cleaning protective coating and painting of equipment, components and systems that are covered under main equipment / system specifications for the CCPP. The components and/or equipment shall be mechanically and /or chemically cleaned during the following stages of the Contract.

- Cleaning in Supplier Works
- Cleaning before painting and/or corrosion protection (application of prime coat)
- Cleaning before erection and during installation.

Cleaning of fabricated component items shall be carried out after fabrication and final heat treatment or welding at manufacturer's works or at site, as appropriate. No paint shall be applied surfaces within 75 mm of field welded connections. In the case of multi-coat systems, every coat shall be stepped back about 30 mm. A strip size allows varying according to the depth of weld and/or for other reasons. These surfaces shall be coated with a consumable preservative and marked.
Please define the overall component that need not to be painting.
For example as below:
Following list of component are not be painted unless specifically required by technical design drawings.

- Joints and surfaces to be field welded (unless a weldable primer, approved by Purchaser, is used. This will be painted according to related painting system after welding in site.). F
- Surfaces in contact with concrete or grout
- Milled and other machine-finished surfaces
- Hot dip galvanized surfaces, grating, chequered/diamond plate, toe plate, handrail & post, ladder and perforated sheet.
- Non-ferrous surfaces (such as copper, brass, aluminum, etc.) unless specified specifically
- Stainless steel surfaces
- Internal surface of pipes, tubes and valves unless specified specifically
- Plastic and / or plastic coated materials unless specified specifically
- Contact surfaces where connections are made by means of high strength friction grip bolts
- Metal embedded in concrete
- Chromium plated metals
- Rubber belts, skirts, gaskets etc.

For cleaning in workshop and before painting, mechanical cleaning by power tool and scrapping with steel wire brushes shall be adopted to clear the surfaces. However, in certain locations where power tool cleaning cannot be carried out, hand scrapping may be permitted with steel wire brushes and/or abrasive paper. Cleaning with solvents shall be resorted to only in such areas where other methods specified above have not achieved the desired results. Cleaning with solvents shall be adopted only after written approval of the Employer / Purchaser.

Machined surfaces shall be protected during the cleaning operations.

In the event of the surfaces not being cleaned to the Purchaser's satisfaction, such parts of the cleaning procedures or agreed alternatives as are deemed necessary to overcome the deficiencies shall be carried out at the supplier's sole expense.

For reclining small areas, hand cleaning by wire brushing may be permitted.

1.2 Codes and Standards

Painting of equipment shall be carried out as per the Codes indicated below and shall conform to the relevant Codes for the material and workmanship.

The supplier shall arrange, at his own cost, to keep a set of latest edition of these standards.

The following codes and standards shall be followed for the surface preparation, surface protection and painting works.

BS EN ISO 4618:2014		Paints and varnishes. Terms and definitions
BS 3083:1988		Specification for hot-dip zinc coated and hot-dip aluminum/zinc coated corrugated steel sheets for general purpose
BS 5252:1976		Framework for color co-ordination for building purposes
BS EN ISO 8501		Preparation of steel substrates before application of paints and related products Visual assessment of surface cleanliness"
BS EN ISO 8502		Test for the assessment of surface cleanliness Preparation of steel substrates before application of paints and related products
BS EN ISO 8503-3		Preparation of steel substrates before application of paints and related products. Surface roughness characteristics of blast-cleaned steel substrates
BS EN ISO 8504		Preparation of steel substrates before application of paints and related products – surface preparation method
BS 7079		General introduction to standards for preparation of steel substrates before application of paints and related products
DIN 8567		Surface Preparation of Metal Surfaces For Thermal Spraying
ISO 2063 -1:2017		Thermal spraying — Zinc, aluminium and their alloys — Part 1: Design considerations and quality requirements for corrosion protection systems
ISO 4624		Paints and Varnishes – Pull off test for adhesion
ISO 2409		Paints and varnishes – Cross cut test
ISO 12944		Corrosion protection of steel structures by protective paint systems
ISO 1461		Hot dip galvanizer coatings on fabricated iron and steel articles – Specification and test methods
SSPC-PA2		Procedure for determining conformance to dry coating thickness requirements
S1S-055900-1967		Surface preparation standards for painting steel surfaces. (Swedish standard Institution)
SSPC-SP 3, SP6 & SP10		Preparation Specifications (Steel structures painting council, U.S.A.).
NACE		National Association of Corrosion Engineers, U.S.A. (NACE).
BS 1710		Specification for identification of pipelines and services
ASTM D4752		Standard test method for measuring MEK resistance Ethyl Silicate (Inorganic) Zinc- Rich Primers by Solvent Rub.
D1212		Standard Test Methods for Measurement of Wet Film Thickness of Organic Coatings
D2200		Standard Practice for Use of Pictorial Surface Preparation Standards and Guides for Painting Steel Surfaces
D3359		Standard Test Methods for Rating Adhesion by Tape Test
D4417		Standard Test Methods for Filed Measurement Of Surface Profile of Blast Cleaned Steel

D4541		Standard Test Method for Pull Off Strength Coating Using Portable Adhesion Testers
D7091		Standard Practice for Non Destructive Measurement of Dry Film Thickness of Non Magnetic Coatings Applied to Ferrous Metals and Non Magnetic ,Non Conductive Coatings Applied to Non-Ferrous Metals
E 376		Standard Practice For Measuring Coating Thickness By Magnetic – Filed or Eddy – Current (Electromagnetic) Testing Methods

1.3 Scope of Work and General Requirements

This specification covers the surface preparation, method of application and material to be used for all coating of equipment, steel structures and piping. Steel material subjected to surface preparation on shop/site shall have minimum requirements in accordance with Rust Grade B (SSPC/SSPM Volume-2).

According to ISO 12944, the plant corrosive level is C5 for outdoor and C3 for Indoor. The outdoor painting shall be resistant to sand/dust storm conditions. However for outdoor Transformers, C4-H will be provided.

Coating materials according to SSPC, EN ISO, ASTM, BIS or DIN standards, shall be used. The paint shall comply with applicable laws, regulations, ordinances etc., of the local authority, state or the nation pertains to the work. The materials shall be matched with each other so that they are compatible. Coatings deviating this specification shall be subject to approval.

The durability of the painting shall be classified as high durability (15 -25 years) as per ISO 12944 and atmospheric corrosivity category C5 (Outdoor) and C3 (Indoor) as per ISO 12944.
The paint to be applied shall be approved by Employer/Purchaser.

All paints & paint material used shall be procured from approved manufacturers. Paint shall be supplied in manufacturers original containers with the description of content, specification No., color, ref no, date of manufacture, shelf-life expiry date & pot life.

The paint manufacturers shall provide coating system data sheet for each coating system to be used containing the following information

a. Surface preparations
b. Film thickness (min and max)
c. Min and max recoating intervals at relevant temperatures
d. Mixing ratio, thinner details and coating repair systems

The sample for testing the paint being used may be taken by the Employer/Purchaser at any time.

In general Shop fabricated equipment shall be delivered to the site coated with a shop applied system or the manufacturer's standard finish in accordance with the requirements of this specification.

For equipment that has received shop prime coat, all touch-up prime coat and additional coats shall be applied in accordance with the coating schedule. It is responsibility of the vendor to ensure compatibility between shop and field applied paint systems.

Necessary precautions shall be provided to all equipment, structures to protect other surfaces from abrasive blasting, coating over spray and spatter. Damage to other surfaces or equipment shall be repaired by the vendor.

The Supplier shall submit the following for review and approval by the Employer/Purchaser's Engineer:

a. Manufacturer's recommended paint scheme for the project
b. Latest published product & instructions for application data,
c. Procedures for surface preparation and application.
d. Pre qualification for equipment and blasting materials, product, procedure and personnel qualifications for the paint and painting systems.
e. Painting repair procedures

Painting records shall contain:

- Equipment/components/location painted
- Date of painting
- Paint details such as specification No, color, date of manufacture, shelf life, expiry date
- Application equipment
- Ambient conditions at the time of painting
- Surface temperature
- Drying time between coating, DFT and number of coatings
- Appropriate work plan for painting.

The supply of all necessary equipment, weather protection, and scaffolding for painting to ensure work is carried out in accordance with the specification and agreed programme.

Maintenance of the paint work until completion of the contract, this shall include repair of any damaged areas caused by third party.

Disposal of painting waste resulting from painting, shall comply with applicable laws, regulations, ordinances etc., of the local authority, state or the nation pertains to the work and coating materials.

It is a mandatory requirement that all operatives working to this procedure take full cognizance and implement necessary safety precautions.

Paint materials that were used in the shop painting (primer, intermediate, and final) including thinner shall be supplied for site touch up by each supplier according to ratio defined in below. Ratio (%) shall be based on the paint consumption quantities that were used at shop.

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- Stack (interior and exterior surfaces): 5%

- Structural steel, Stair tower, Inlet duct & outlet duct and Casing: 5%

- For Hot spot repair on duct and casing (by duct and casing supplier): sufficient quantities (Inorganic zinc primer and Silicone aluminum paint)

- Structural skid steel (Incl. saddle or leg): 5%

- Pipe and Pipe hanger & Support: 5%

- Silencer, Valve wheel: sufficient quantities

- Tanks: 5%

- Canopy, Crane, Hoist: 5%

- Hot dip galvanized items: 5% (Cold galvanizing paint)

- Any Other Items/Elements/Equipment: 5%

1.4 Cleaning at manufacturer's works

Mechanical cleaning shall preferably be carried out by abrasive blasting. The Supplier is prepared to consider alternative methods such as chemical cleaning provided they achieve the necessary surface condition.

In case of chemical cleaning, the detailed procedure for chemical cleaning as well as the system for which chemical cleaning is required shall be submitted by the supplier for Purchaser's approval. The procedure shall comprise of pre-treatment and acid treatment to achieve cleanliness equivalent to that specified for mechanical cleaning.

Surface condition:

The Metal surfaces shall be clean and free of mill scale, rust, dirt, grease and any other deleterious matter.

Where metal surfaces are to be painted, the surface profiles shall conform to the painting specification requirements.

Where this does not apply, surfaces shall have a surface texture not coarser than Grade 80 abrasive paper.

Abrasives:
Abrasive should be free from detrimental amount of water-soluble, solvent-soluble, acid soluble or other soluble contaminants. Criteria for selecting and evaluating abrasives are given in SSPCAB1,SSPC-AB2, SSPC-AB3 must be considered.

Abrasives containing silica, silicates or slag residues shall not be used for water/steam side surfaces of plant except for cleaning sand castings, where hydro blasting may be employed.

For austenitic materials only, abrasives containing 98% or more of alumina, $Al_2 O_3$, shall be used.

Removal of abrasive and debris:

After cleaning, abrasive and debris shall be thoroughly removed for components.

1.5 Protection at manufacturer's works

As soon as all items have been cleaned and within four hours of the subsequent drying, they shall be given suitable anti-corrosion protection.

All water, air and steam side surfaces shall be protected by the application of approved water soluble corrosion inhibitors, or vapor phase inhibitors that can be subsequently removed by site water washing or steam blowing.

The gas side of steam generating plant items shall be protected by the application of temporary protective that do not require to be removed before commissioning, but which are removed during initial firing.

The rate of application of volatile corrosion inhibitors shall be at least 10 grams per square meter or 35 grams per cubic metre, whichever is the greater, except for pipes up to 300 mm diameter for which the minimum application rates shall be 5 grams per square metre.

Immediately after the protective treatment has been applied all vessels and pipes shall be suitably sealed off by discs or caps or approved alternatives to prevent ingress from the surroundings.

Cylindrical plugs shall not be driven into the ends of pipes. These protective covers shall not be removed until immediately before final connection is made to the associated equipment.

1.6 Weather conditions

a. If the below conditions are prevalent, painting shall be delayed or postponed until conditions are favorable
b. When the surrounding air temperature or the temperature of the surface to be painted is below 5°C; (Subjected to lower drying or curing limit of coating applied)
c. When the surrounding air temperature or the temperature of the surface to be painted is higher than paint manufacturers specified maximum limit for application;
d. When surfaces are wet or damp, or in rain, snow, fog, or mist;
e. When the temperature is less than 3°C above the dew point;
f. When it is expected that the air temperature will drop below 5°C or less than 3°C above the dew point within eight hours after application of the painting.

If during the painting work, low temperatures, excessive humidity, mist or rain occur, the paint, which has already been applied, must first dry by eliminating the above mentioned conditions. Places, which have been damaged, must then be removed and renewed.

Wind speed and direction should be monitored during outdoor painting applications to avoid dry spray and overspray of paint. If required base on wind speed and direction, stop painting or employ some measure to prevent overspray.

1.7 Surface preparation

In preparing any surface to be coated, all loose paint, dirt, grease, rust, scale, weld slag or spatter or any other extraneous material shall be removed and defects repaired, so as to obtain a clean, dry, even surface to receive the priming or finishing coat (s) as called for in the painting schedules. Sharp edges should be rounded, especially when tank linings have to be applied.

All machined surfaces, including flange faces, shall be suitably covered to prevent damage during surface preparation.

All corners and sharp edges should be rounded to a 2mm radius, especially when tank linings have to be applied. For internal coatings subject to immersion conditions, assure that weld surfaces conform to NACE SP0178 Weld Preparation Designation “C” unless otherwise specified.

All surfaces should be blast cleaned whenever possible.

Surface preparation methods:

Bare steel surfaces should be prepared by one of the methods described below in order of preference and in accordance with ISO 8501.

If the relative humidity is greater than 80%, blast immediately prior to painting. Blast cleaning shall not be conducted when the temperature of the surface is less than 3°C above dew point of surrounding air. Surface shall not be left unpainted overnight.

a. White metal blast cleaning Sa 3 or SSPC - SP 5

Sa 3 Blast cleaning to bare metal. Mill scale, rust and foreign matter must be removed completely. Subsequently, the surface is cleaned with vacuum cleaner, clean dry compressed air or a clean brush. It must then have a uniform metallic color and correspond in appearance to the prints designated Sa 3.

b. Near white metal blast cleaning Sa 2 1/2 or SSPC - SP 10

Sa 2 1/2. very thorough blast cleaning. Mill scale, rust and foreign matter shall be removed to the extent that the only traces remaining are slight imperfections in the form of spots or stripes.

Subsequently, the surface is cleaned with a vacuum cleaner, clean dry compressed air or a clean brush. It must then correspond in appearance to the prints designated Sa 2 1/2.

Mechanical cleaning should only be used when procedures (a) and (b) are not practicable.

c. Near white metal blast cleaning Sa 2 DIN EN ISO 12944-1

Thorough blast cleaning. The jet is passed over the surface long enough to remove all mill scale and rust and practically all foreign matter. Finally, the surface is cleaned with a vacuum cleaner, clean and dry compressed air or a clean brush. It should be grayish in colour. It must then correspond in appearance to the prints designated Sa 2.

d. Very thorough hand and power tool cleaning St 3

St 3 very thorough scraping, wire-brushing, machine brushing - grinding etc. are to be preferred. Surface preparation as for St 2, but much more thoroughly. After the removal of dust, the surface must have a pronounced metallic sheen and correspond to the prints designated St. 3.

e. Thorough hand and power tool cleaning St 2

St 2 Thorough scraping, wire-brushing, machine brushing, grinding etc. The treatment shall remove loose mill scale, rust and foreign matter. Subsequently, the surface is cleaned with a vacuum cleaner, clean dry compressed air or a clean brush. It should then have a faint metallic sheen. The appearance must correspond to the prints designated St 2.

f. Air Blasting with Non-Metallic Abrasives Powder

Whenever the "Duplex"-process is to be applied (hot dip galvanising followed by painting), prepare the hot dip galvanised surface by water washing to remove flux residues and careful air blasting with non-metallic abrasive powder. Use an abrasive with grain size from 0.1 to 0.5 mm, at a greatly reduced air pressure, max. 2 bar (g) (28 psig).

This procedure also applies to stainless steel and aluminium surfaces to be coated.

Surface preparation methods	**SIS 055900**	**DIN EN ISO 12944Part-4**	**BS 4232 only for blasting**	**SSPC-Vis**
Blasting acc to item (a),(b),(c),	Sa 3		First quality	White metal SP 5
Blasting acc to item (b)	Sa 2 1/2		Second quality	near White SP 10
Blasting acc to item (c)	Sa 2		Third quality	Commercial blast SP 6
Hand/or power tool derusting acc to item (e)	St 2		--	Hand tool cleaning SP 2
acc to items (d) and (e)	St 3		--	Power tool cleaning SP 3
Flame jet cleaning		FI	--	Flame cleaning SP 4
Pickling		Be	--	Pickling

Steel structures to be blast cleaned have to be free of pitting and other severely corroded places in accordance with BS 7079 and ISO 8501.

The abrasives used for blast-cleaning shall be graded flint, grit, shot or silica sand and shall be such that they will produce an average keying profile on the blast-cleaned surface of not more than 40 microns. Where metal surfaces are to be painted the surface profiles shall be “medium (G)” or “medium (S)” as defined in ISO 8503-1 or as per Paint Manufacture recommended surface profile for better performance of protective painting system.

An air pressure of 7 bar g at the nozzle shall be used.

After blast-cleaning, all accumulated grit, dust, etc., must be removed leaving the surface clean, dry and free of mill scale, rust grease and other foreign matter.

In the event of rusting after completion of the surface preparation, the surface must be cleaned again in the manner specified.

Oil, grease, soil, cement, salts, acids or other corrosive chemicals shall be cleaned from steel surfaces, by the use of solvents, emulsions or cleaning compounds. The final wiping shall be with clean solvent and clean rags or brushes. There shall be no detrimental residue left on the surface.

Primed areas which suffer damage must be spot blasted on site to a degree of cleanliness Sa 2 1/2 before, touching up.

Protective coating must be applied as quickly as possible after the completion of surface preparation no matter what cleaning method has been used.

No blast-cleaned surface shall be allowed to remain uncoated overnight.

Steel work protected by shop primer after arrival on site must be cleaned of salt, sand, oil etc. Before the first coat of paint is applied on site. Shop primer damaged during transport must be rectified by blast-cleaning and coating before application of the site coats.

Wood surfaces shall be sanded clean. All nail holes shall be puttied and sanded before priming.

Concrete: If a protective coating is required, concrete shall be allowed to cure before painting.

1.8 Preparation of coating materials

All containers shall remain un-opened until required for use. Mix and prepare painting material according to the manufacturer’s directions.

Primers and paints which have livered, gelled or otherwise deteriorated shall not be used.

The oldest primer or paint of each kind shall be used first.

All ingredients in any container shall be thoroughly mixed before use, and shall be agitated frequently during application to keep the primer in suspension.

Primer or paint mixed in the original container shall not be transferred until all settled pigment is incorporated into the body of liquid.

Mixing in open containers shall be done in a well ventilated area. Stir material before application to produce a mixture of uniform density and stir as required during application of material. Do not stir surface film into the material. Remove the film and, if necessary, strain the material before using.

Primer or paint shall be mixed in a manner ensuring the breakdown of all lumps, complete dispersion of pigment and uniform composition.

Two-component primers shall be mixed in accordance with the manufacturer's instructions.

Thinners shall not be added to primers or paints unless necessary for proper application according to the manufacturer's instructions.

When use of thinners is permitted, it must be added to the primer or paint during mixing.

1.8.1 Primer Paint

After the surface is prepared, one coat of suitable primer shall be applied. After this first coat is dried up completely, second coat of primer shall be applied.

Primer shall be applied by brushing to ensure a continuous film without 'holidays'. The dry film thickness of each coat shall be as specified in the Annexure- 2 -Paint System of this specification.

The primer should be worked by brush application or airless spray to cover the crevices, corners, sharp edges etc. in the presence of inspector.

The shades of successive coats should be slightly different in color in order to ensure application of individual coats, the thickness of each coat and complete coverage should be checked as per specification approved by Contractor before application of successive coats.

The supplier shall provide standard thickness measurement instrument with appropriate range(s) for measuring.

Elko meter for measuring the Dry film thickness of each coat, surface profile gauge for checking of surface profile in case of sand blasting. Holiday detectors and pinhole detectors for checking the painted surface discontinuities should be provided by the Supplier.

The supplier shall make arrangements for paint manufacturer to provide expert technical service at site as and when required free of cost and without any obligation to the Employer/Purchaser, as it would be in the interest of the manufacturer to ensure that both surface preparation and application are carried out as per their recommendations.

Final inspection shall include measurement of paint dry film thickness, check of finish and workmanship.

1.8.2 Rub down and Touch Up of Primer

The shop coated surfaces shall be rubbed down thoroughly with emery paper to remove all dust, rust and other foreign matters, washed, degreased, then cleaned with warm fresh water and air dried.

The portions, from where the shop coat has peeled off, shall be touched up and allowed to dry before applying a coat of primer.

The compatibility between shop coat and field primer shall be ascertained from the paint manufacturer. In case degreasing with white spirit is not effective, the surface shall be finally wiped clean with aromatic solvent like xylol or light naphtha.

1.8.3 Touch-Up Painting

a) Shop or field touch-up painting shall be the same as the specified painting material and color as used for the original painting system. The touch-up painting shall be applied to the damaged surface after completion of specified surface preparation and shall meet painting system manufacturers recommendations regarding minimum surface preparation methods, surface profile, and touch up and "re-coat" times.

b) Vendor shall ship a sufficient quantity of paint, same as used for the shop applied coats, to the project site for use as touch-up paint.

c) Touch-up painting shall overlap the original coat by not less than 2.5 cm to ensure continuity of the painting.

d) For surfaces which have received two coats, touch-up painting shall consist of two coats also; such two-coat touchup shall conform to the same requirements as for two-coat painting work.

e) Field touch-up painting shall be applied to the following surfaces:

  i. Surfaces where shop or field applied painting has been marred, scratched or otherwise damaged, due to shipping, handling, erection, installation, weathering, etc.
  ii. Heads of the field bolts and nuts, and adjacent surfaces left uncoated in the shop.
  iii. Surfaces of field welds and adjacent surfaces left uncoated in the shop and where painting was damaged during the welding process.
  iv. Surfaces of ferrous fasteners not otherwise protected.
  v. Exposed fabrication, erection or shipping marks shall be cleaned off and the areas touch-up coated to match the adjacent surfaces.

1.8.4 Non Compatible Shop Coat Primer

a) The compatibility of finishing coat shall be confirmed from the paint manufacturer. In the event of use of primer such as zinc rich epoxy, inorganic zinc silicate etc., the paint system shall depend on condition of shop coat. If the shop coat is in satisfactory condition showing no major defect, the shop coat shall not be removed. The touch up primer and finishing coat(s) shall be identified for application by Purchaser. Shop coated (coated with primer & finishing coat) equipment shall not be repainted unless paint is damaged.

b) Shop primed equipment and surfaces shall only be 'spot cleaned' in damaged areas by means of power tool brush cleaning or hand tool cleaning and then spot primed before applying one coat of field primer unless otherwise specified. If shop primer is not compatible with field primer then shop coated primer shall be completely removed before application of selected paint system for particular environment. For package units/equipment, shop primer shall be as per the paint system given for particular environment.

c) In case of existing paint, compatibility between finishing coat and new selected finish coat shall be ascertained before application of finish coat. In case, the coat is selected for upgrading existing alkyd coating to high performance coating then, surface preparation shall be by manual/mechanical means to remove loose rust, peeled off/damaged paint, but sound old coating need not be removed. It shall be touched with suitable primer wherever it has peeled of before application of tie coat. The tie coat shall be applied after 7 days of curing of the primer. If, new paint system is not suitable to upgrade existing coating then complete paint shall be removed by mechanical or blast cleaning before application of new coating system.

1.8.5 Finish Paint

Suitable Finish paints as per the schedule shall be applied for the jobs. The color/shade shall be as approved by the Employer/Purchaser. After cleaning the dust on the dried up primer, first coat of finished paint shall be applied. After this first coat dries up hard, the surface is wet scrubbed cutting down to a smooth finish and ensuring that at no place the first coat is completely removed. After applying second coat, allowing the water to get evaporated completely, third finish coat of finish paint may be applied(if applicable).

1.9 Paint Materials

The paints shall conform to the specifications given in this Annexure and class - 1 quality in the products range of any of the reputed international manufacturers.

1.10 Storage

All paints and painting material shall be stored only in rooms to Contractor’s approval. All necessary precautions shall be taken to prevent fire. The storage building shall preferably be separated from adjacent buildings. A signboard bearing the words "PAINT STORAGE - NO NAKED LIGHT - HIGHLY INFLAMMABLE - DANGER - NO SMOKING" shall be clearly displayed outside. All paints shall be stored in the safest manner so that no container rolls down and causes accidents. The shelf life of the paints shall be ensured so that the paint materials are not in storage and use after the date of expiry.

1.11 Application

Health and safety of work

The supplier has to check all painting work to be carried out according to the specification of the paint supplier further to all relevant prescriptions and regulations concerning the health and safety of work.

The paint supplier has to present a written specification including at least the flash point of the paints, ventilation requirements, handling precautions such as inhalation, eye and skin protection, and first aid procedure, storage requirements, spill or leak procedure, fire precaution, waste disposal.

Methods

Apply painting materials in accordance with the paint manufacturer’s printed instructions. Use applicators and techniques best suited for the type of material being applied. Method of application should be selected as per recommended by Paint Manufacturer’s.

Temporary corrosion protections are to be completely removed prior to applying the definite one, in acc. with DIN EN ISO 12944-1.

All prime coatings shall be applied by brush or airless spray or a combination of these methods, as approved by the coating manufacturer.

Paintings must be allowed to cure according to the relevant Product Technical Data Sheets before being recoated or placed into service. Drying time requirements recommended by the paint manufacturer should be followed exactly. Always refer to the relevant Product Technical Data Sheets and working procedures for paint material.

All doors, windows, stairways, handrails (if painted), bolts, flanges and equipment supports shall be finish painted by brush. Application of paint shall be applied evenly, free of runs or sags, drips, inclusions or other defects, with no evidence of poor workmanship. Finish surfaces shall be free from defects, holidays or blemishes. All deficiencies and defects should be corrected.

Spray guns should not be used outside in windy weather or near surfaces of a contrasting colour unless the latter is properly protected.

All cold-spray painting shall be done using standard equipment in accordance with accepted standards and methods.

Care has to be taken not to connect spraying devices for nitro and backelite paints simultaneously to oil based paints.

Clean Up

Unused materials, discarded materials, spoiled materials, rags, empty paint material containers, etc. shall be cleaned up, properly stored and removed from the job site. No debris, trash or other such materials may be allowed to accumulate.

Faying Surface Paintings

a. Paintings and their overspray shall NOT be applied to the faying surfaces of bolted connections, unless it has been certified by the manufacturer to meet the requirements in AISC "Specification for Structural Joint Using ASTM A325 or A490 Bolts" for the connection design class or as per EN code.
b. Paintings, including inadvertent overspray, shall be excluded from areas closer than one bolt diameter but not less than one inch (25 mm) from the edge of the hole and areas within the bolt pattern on structural steel faying surfaces.
Paint applied to items that are not being painted shall be removed at the supplier's expense, leaving the surface clean, unstained and undamaged.

Dry film thickness (DFT)

To the maximum extent practicable the coats shall be applied as a continuous film of uniform thickness and free of pores. Overspray, skips, runs, sags and drips should be avoided. The different coats shall not be of the same color.

For a composite paint or coating system consisting of several coats, the total DFT must be at least equal to the sum of the minimal DFT's for the individual coats. If, the paint system does not have the required minimum DFT those areas should be marked & repainted. If the occurrence of those areas is high, the complete surface must be repainted. It is also critically important to check the DFT of primers and intermediate coats and to correct them where necessary.

For paintings based on Zinc silicate, the DFT is limited as well on minimum DFT as on maximum (150µm) because of the risk of mud cracking. It is recommended for two pack inorganic zinc rich ethyl silicate primers that one of the subsequent coats be used as a tie coat.

Consumption of paints

Has to be evaluated according to DIN 53220. The paints shall be tested as per – ISO: 3668-2017, ISO- 15528.

Each coat of paint shall be allowed to harden before the next is applied. For epoxy paint the hardening time normally is 12-14 hours. Suppliers' recommendations regarding hardening time of epoxy paints must be followed.

Particular attention must be paid to full film thickness at edges.

The minimum total dry film thickness of the paint systems shall be as recommended in the following tables below. The DFT is given in microns (millionths of a metre).

1.12 Galvanizing

Galvanizing works shall conform in all respect to BS EN ISO 1461, BS 3083 whatever requires the higher quality and shall be performed by the hot dip process, unless otherwise specified.

It is essential that details of steel members and assemblies which are to be hot-dip galvanized should be designed in accordance with BS 4479.

Vent-holes and drain-holes should be provided to avoid high internal pressures and air-locks during immersion, which may cause explosions, and to ensure that molten zinc is not retained in pockets during withdrawal.

Careful cleaning of welds is necessary before welded assemblies are dipped. The welds and the surrounding metal should be cleaned separately, preferably be blast-cleaning, because the usual preliminary pickling cannot be relied on to remove the welding slag.

All defects of the steel surface including cracks, surface laminations, laps and folds shall be removed in accordance with BS. 4360. All drilling, cutting, welding, forming and final fabrication of unit members and assemblies shall be completed, where feasible, before the structures are galvanized. The surface of the steelwork to be galvanized shall be free from paint, oil, grease and similar contaminants in accordance with DIN EN ISO 12944, part 4 and BS EN ISO 1461. The weight of zinc coating per unit area has to be noted in the manufacturing documents in accordance with BS EN ISO 1461

The minimum average coating weight shall be as specified in Table 1 of BS EN ISO 1461

After fixing, bolt heads, washers and nuts shall receive two coats of zinc-rich paint. Connections between galvanized surfaces and copper, copper alloy or aluminum surfaces shall be protected by suitable preferably hydrophobe tape wrappings to the Purchaser's approval.

1.13 Sprayed Metal Coatings

Corrosion protection may be also achieved by spraying of suitable metals as zinc and/or aluminum on the surfaces of structures. For special cases tin, copper, lead can be used as well. Methods of surface preparation have to conform to BS EN ISO 2063-1or to DIN 8567. A proper treatment of the surface followed by an immediate spraying is to apply to ensure adhesion of the sprayed metal. The surface has to be clean, free of impurities, rust, mill scale and rough enough to have binding properties to ensure good enticulation with the sprayed layer. Suitable roughness can be achieved by blast cleaning acc. to BS 7079 or DIN 8567. Welds are to be cleaned and prepared with special care. All surfaces to be treated have to be dry and accessible.

Application of coatings, requirements for thickness, adhesion, composition of coating metals, and subsequent treatment have to conform to BS 2569, DIN 8565 and 8567.

Testing of the spray coated layers are to be carried out in accordance with DIN 8565.
The Supplier has to specify the type, composition and thickness of the sprayed metal and of the sealing coating according to DIN 8565 including the corresponding warranties and tests if, sprayed metal coating will be applied.

Safety of work:

All precautions connected with this type of application of corrosion protection have to be in accordance with German regulation DVS 2307, page 1. 2.

Sprayed, unfused coating of metals and metallic compounds applied by combustion gas flame, plasma arc, detonation and similar processes, and the preparation of components, spraying techniques, sealing, finishing and inspection shall be according to BS EN 13507.

The hot galvanized surface has to be cleaned before the application of the coats to remove corrosion products, dirt, dust, grease.

The cleaning can be achieved by

- brush off
- washing with 1 - 1.5 % ammonia water with up to 0.1 % detergent added and followed by wet grinding to turn the foam to grey color,
- steam blasting.

1.14 Warning Notes / Signals

This Instruction serves the identification of the coated surfaces that are received from shop in assembled condition / module wise.

The warning note shall prevent any possible damage to the coated surfaces during transportation / assembly at site.

Eg.: Welding work OR Heat treatment work on the outside of coated or lined surfaces is prohibited.

1.15 Colour Code for Piping

a. The color code scheme is intended for identification of the individual group of the pipeline. The system of color coding consists of a ground color and color bands superimposed on it. The color coding for the identification of pipelines shall comply with **Annexure – 1** of this specification.

Ground Color shall be applied throughout the entire length for uninsulated pipes. For insulated pipes, on the metal cladding or on the pipes of material such as non-ferrous metals, austenitic stainless steel etc., ground color coating of minimum 2m length or of adequate length not to be mistaken as color band shall be applied at places requiring color bands. Color band(s) shall be applied at the following location.

i. At battery limit points

ii. Intersection points & change of direction points in piping ways.

iii. Other points, such as midway of each piping way, near valves, junction joints of service appliances, walls, on either side of pipe culverts.

iv. For long stretch/yard piping at 50m interval.

v. At start and terminating points.

b. Flow direction shall be indicated by an arrow in the location stated above and as directed by Purchaser. Colors of arrows shall be black or white and in contrast to the color on which they are superimposed. The size of the arrows shall confirm to BS-1710 (2014) / ASA A13.1 (2020). Product names shall be marked at pump inlet, outlet and battery limit in a suitable size as approved by Purchaser. As a rule minimum width of color band shall conform to 75 mm up to 300 NB and to 100 mm over 350 NB. Whenever it is required by the Purchaser to indicate that a pipeline carries a hazardous material, a hazard marking of diagonal stripes of red and golden yellow as per BS-1710 (2014) / ASA A13.1 (2020) shall be painted on the ground color.

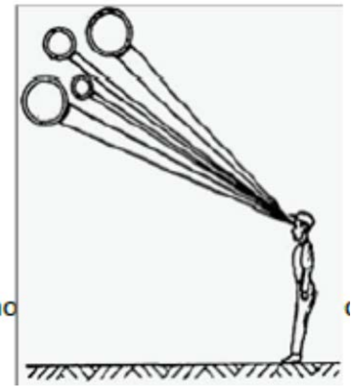
c. For Closed circuits system where it is necessary to indicate separately, the flow and return pipes, this shall be done by the use of the word 'FLOW' or the letter 'F' on the one pipe and the word 'RETURN' or the letter 'R' on the other.

d. All uninsulated piping systems and supports shall have two coats of suitable primer coats and with suitable finish paints as per Annexure-2 Painting system. Service of the pipe/line designations shall be painted on all pipes at visible locations.

1.16 Identification of Vessels, Piping etc.

Equipment number shall be stenciled in black or white on each vessel, column, equipment and machinery after painting.

Line number in black or white shall be stenciled on all the pipelines of more than one location as directed by Contractor; size of letters printed shall be 150 mm (high) for column & vessels. 50 mm (high) for pump compressor and other machinery and shall be as per BS 1710/ ANSI A 13.1 for piping. The storage tanks shall be marked as detailed in the respective drawing.
Attention shall be given to visibility with reference to pipe markings. Where pipelines are located above or below the normal line of vision, the lettering (Legend) shall be placed below or above the horizontal centerline of the pipe (see figure).



1.17 Inspection and Testing

a) All painting materials including primers and thinners brought to site for application shall be procured directly from manufacturer as per specifications and shall be accompanied by manufacturer's test certificates. Paint formulations without certificates are not acceptable. Contractor at his discretion, may call for tests for paint formulations. Supplier shall arrange to have such tests performed including batch wise test of wet paints for physical & chemical analysis. All costs thereof shall be borne by the Supplier. The paints shall be tested as per ISO:15528 / equivalent international standard and approved by the Employer/Purchaser.

b) The painting work shall be subject to inspection by Employer/Purchaser at all times. In particular, following stage wise inspection shall be performed and supplier shall offer the work for inspection and approval of every stage before proceeding with the next stage. The record of inspection shall be maintained in the registers. Stages of inspection shall be surface preparation, primer application and each coat of paint. In addition to above, record shall include type of shop primer already applied on equipment e.g. red oxide zinc chromate or zinc phosphate or Silicate primer etc.

c) Any defect noticed during the various stages of inspection shall be rectified by the Supplier to the entire satisfaction of Purcahser before proceeding further. Irrespective of the inspection, repair and approval at intermediate stages of work, supplier shall be responsible for making good of any defects found during final inspection/guarantee period/defect liability period as defined in general condition of contract. Dry film thickness (DFT) shall be checked and recorded after application of each coat and extra coat of paint shall be applied to make-up the DFT specified without any extra cost to the Purchaser.

1.18 Guarantee

The supplier shall guarantee that the chemical and physical properties of paint materials used are in accordance with the specifications contained herein/to be provided during execution of work. The supplier shall produce test reports from the manufacturer regarding the quality of the particular batch of paint supplied. The purchaser shall have the right to test wet samples of paint at random for quality of the same. Batch test reports of the manufacturer's for each batch of paints supplied shall be made available by the supplier.

1.19 Buried Pipe Protection

Purpose

a) This section to provide minimum requirements and information on the equipment, material and methods utilized for protective coatings and field wrapping of steel and other metallic pipe and fittings.
b) Each segment of metallic pipe, new or replacement shall have a protective coating.

Field wrapping

The cleaning and application of plastic tape for use in the buried uninsulated piping system is essential for corrosion protection.
The following products are acceptable for field-installed:

a) For pipe, weld joint and transition fittings, primer and cold tape or approved equivalent products may be utilized.

b) For irregular shaped fittings and valves, Trenton Temcoat grease and Trenton polyply wrap or approved equivalent products may be utilized.

Each product shall be applied in accordance with the manufacturer’s instructions.

Application directions

a) Surfaces to be wrapped must be clean and dry. Any oil or grease must be removed.
b) Any burrs or sharp edges must be removed or smoothed.
c) Before wrapping application, all bare pipe and fittings to be wrapped must be coated with required adhesive material.
d) Weld beads shall be covered with one wrap or layer of tape prior to spiral wrapping.
e) Tape shall be spirally wrapped accordance with the following “Table 1”.

Note

i. When field wrapping a transition to PE, neither primer nor tape should be applied within ¼” of the PE. Electrical tape should be applied to the remainder of the fitting. All transition fittings after welding will have the entire steel portion wrapped.
ii. The first column is the nominal size of the pipe. The third, fourth and fifth columns tell your requirements per 100 lineal feet of pipe, using a spiral wrap with both a 1" and a 50% over lap.
iii. Be sure to note that as per common industry standards the length of the roll changes for the 6" wide tape. Double check your product before going to the job site as many manufacturers vary between 50 and 100 foot standards, regardless of the width.

Table 1

Nominal Pipe Size in Inches	Surface Square Feet per 100 of Pipe	Rolls of 2"x 100' Tape	Rolls of 4" x 100' Tape		Rolls of 6" x 50 ft Tape	
		1" Overlap	1" Overlap	50% Overlap	1" Overlap	50% Overlap
½	22	2.7				
¾	27.5	3.3				
1	34.4	4.2				
1 ¼	43.5	5.2	1.8	2.6		
1 ½	49.7	6	2	3		
2	62.2		2.5	3.8		
2 ½	75.3		2.7	4.5		
3	91.6		3.7	5.5		
4	117.8		4.8	7.1	5.7	9.5
5	145.6		5.9	8.8	7	11.7
6	173.4		7	10.4	8.4	13.9
8	225.8		9	13.6	10.9	18
10	281.4		11.3	16.9	13.5	22.5
12	333.8		13.4	20	16	26.7

ANNEXURE-1 STANDARD FINAL COLOUR OF EQUIPMENT AND PIPING

1.0.0 STANDARD COLOR CODE FOR MECHANICAL EQUIPMENT

Item. no	Description	Colour	RAL Code	Remarks
1.0	GTG and Its Auxiliaries	Supplier's Standard		
2.0	STG and Its Auxiliaries	Supplier's Standard		
3.0	HRSG and Auxiliaries	Supplier's Standard		
4.0	Bypass Stack	Supplier's Standard		Stack Painting Scheme to be painted according to ICAO.
5.0	General indoor equipment, such as pumps, filter, strainer	As the associated piping (unless standard equipment)		
6.0	General outdoor equipment, such as Air Cooled Condenser, pumps, vessels, filter, strainer etc.	As the associated piping (unless standard equipment)		
7.0	Crane bridges, trolleys, hoist, etc., and other mobile equipment	Melon Yellow	RAL 1028	
8.0	Tanks Raw water tank Treated water tank Maintenance oil tank New oil tank Used oil tank Naphtha storage tank Light fuel oil (Gasoil/Diesel) storage tank	 Pastel Blue Light Blue Colza Yellow Colza Yellow Colza Yellow Ochre Brown Red Lilac	 RAL 5024 RAL 5012 RAL 1021 RAL 1021 RAL 1021 RAL 8001 RAL 4001	For raw water / fire water tank, color will be RAL5024 and there will be red band on the fire fighting water level.
9.0	Grating, tread, toe plate, handrail and post, ladder, chequered plate	Hot dip Galvanized	-	
10.0	Structural steel (Interior & Exterior) Pipe supports and aux. steel structures,	Silver Grey	RAL 7001	Chimney Structure RAL 5019 (Capri Blue)

Item. no	Description	Colour	RAL Code	Remarks
	equipment base plates Metallic Stairways			
11.0	Pipe hangers and spring case, struts, clamp	Supplier Standard		
12.0	Fire Fighting main equipment	Flame red	RAL 3000	
13.0	C&I Control Panel (Indoor)	Light Grey	RAL 7035	

Notes:

1. This color code basically refers to BS:1710(2014) / ASAS A13.1(2020) for piping with necessary modifications
2. For any item left out, color coding will be decided after Employer/purchaser approval.
3. For mass production of equipment, color code shall be as per manufacturer standard.

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2.0.0 STANDARD COLOR CODE FOR ELECTRICAL EQUIPMENT

Sl.No	Description	Colour	Color No.
1	Generator	OEM standard	
2	Generator circuit breaker a) Outdoor	OEM standard	
3	Transformers	Pebble grey	RAL 7032
4	Bus ducts	Pastel Turquoise	RAL 6034
5	Junction boxes	Manufacturer standard	
6	HT/LT Switchboards, Distribution boards		
a)	Indoor/Outdoor	Manufacturer standard	
7	Control & Relay panels		
b)	Indoor/Outdoor	Manufacturer standard	
8	C&I Control Panel	Manufacturer standard	
9	UPS Panel, charger panels	Manufacturer standard	
10	Black Start Diesel Generator	Manufacturer standard	
11	NGR	Pebble grey	RAL 7032
12	Motor	Capri Blue	RAL 5019
13	Lighting fittings	Manufacturer standard	
14	Cable trays	Galvanized	
15	Modulating and on/off shut off valves	OEM standard	

1. For interior coating, manufacturer's standard can be adopted subject to Employer/Purchaser's approval.
2. For mass production equipment color code shall be as per manufacturer standard.

3.0.0 COLOR CODING FOR IDENTIFICATION OF PIPELINES USED IN POWER PLANTS

Sl. No	Description	Colour	RAL Code	Remarks
1	Steam	Silver	RAL 9018 (Heat Insulated/Grey)	Steam lines will be heat insulated. Labeling color on insulation TBD.
2	Closed Cooling Water	Pastel Blue	RAL 5024	
3	Service Water	Traffic Blue	RAL 5017	
4	Potable Water	Mint Turquoise	RAL 6033	
5	Raw Water	Capri Blue	RAL 5019	
6	Feed Water	Light blue	RAL 5012	
7	Condensate Water	Light blue	RAL 5012	
8	Fuel Gas (NG)	Luminous Yellow	RAL 1026	
9	Gas Oil (LDO)	Red Lilac	RAL 4001	
10	Naphtha (HFO)	Oche Brown	RAL 8001	
11	Fire Water	Flame Red	RAL 3000	For galvanized pipe Flame red will be applied as band.
12	Compressed Air	Pastel Green	RAL 6019	
13	Waste	Signal Black	RAL 9004	
14	Chemical Feed (Phosphate)	Patina Green	RAL 6000	Color Band
15	Chemical Feed (Ammonia)	Pure White	RAL 9010	Color Band
16	Chemical Feed (Oxygen Scavenger)	Stone Grey	RAL 7030	Color Band
17	Lubricating Oil	Colza Yellow	RAL 1021	
18	Hydraulic Oil	Light Brown	RAL 8002	

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Sl. No	Description	Colour	RAL Code	Remarks
19	Nitrogen Gas	Ochre Yellow	RAL 1024	
20	Vacuum /Air evacuation system	Pearl White	RAL 1013	
21	DM water	Ochre Brown	RAL 8001	Color Band
22	CO_2 (GT Fire Fighting)	Flame Red	RAL 3000	CO_2 gas piping used for Fire Fighting. So, that flame red color is used.

Notes

1. Covering capacity and DFT depends on method of application. Covering capacity specified above is theoretical. Allowing the losses during application, minimum specified DFT shall be maintained.

2. All primers and finish coats shall be cold cured and air dried unless otherwise specified.

3. All paints shall conform to relevant Standard and shall be applied in accordance with manufacturer's instructions for surface preparation, intervals, curing and application. The surface preparation, quality and workmanship shall be ensured.

4. Technical data sheets for all paints shall be supplied at the time of submission of quotations.

5. In case of use of epoxy tie coat, manufacturer shall demonstrate satisfactory test for inter coat adhesion. In case of limited availability of epoxy tie coat, alternate system may be used taking into consideration the service requirement of the system.

6. Supplier will submit the final color shade for all equipment & piping under his scope for final approval by Employer/Purchaser as per RAL color code.

4.0.0 COLOR CODING FOR IDENTIFICATION OF FIELD INSTRUMENTS AND VALVES USED IN POWER PLANTS

1. Color coding for field instruments and valves shall be as per OEM Standard.

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ANNEXURE-2: PAINTING SYSTEMS

Cleaning, Protective Coating and Painting. - Systems designed as per ISO 12944 with high Durability of 15 - 25yrs.

Surface/ Location	Paint Number	Temp	Surface prep	Coat	No. of coats	Application	Generic Type	NDFT/ Coat
	PS-1							
Structural Steel work (Indoor- C3)		Less than 100 deg C	SA 2.5	Primer	1	Shop	InOrganic Zinc Silicate or Zinc rich	60
				Intermediate	1	Shop	High Build Epoxy	50
				Finish	1	Shop	Aliphatic Polyurethane	50
							Total	**160**
Structural Steelwork, (Outdoor – C5)	PS-2	Less than 100 deg C	SA 2.5	Primer	1	Shop	InOrganic Zinc Silicate or Zinc rich	60
				Intermediate	1	Shop	High Build Epoxy	200
				Finish	1	Shop	Aliphatic Polyurethane	60
							Total	**320**
Steel Tanks inside Surface (Total) for Gas Oil Storage	PS-3	Less than 50 deg C	SA 2.5	Primer	2	Shop	Zinc Rich Primer	40
				Finish	2	Site	2 Pack Epoxy Mastic	150
							Total	**380**
Underneath of Bottom plate			SA 2	Primer	2	Shop	In Organic Zinc Silicate	40
				Finish	1	Site	FRP Phenolic Epoxy coating	125
							Total	**205**
Steel Tanks outside Surface (Total) for Oil Storage		Less than 50 deg C	SA 2.5	Primer	1	Shop	Two pack inorganic zinc rich ethyl silicate /Two pack zinc rich epoxy paint	60
				Intermediate	1	Site	Two pack high build epoxy polyamide MIO	150
				Finish	1	Site	Two-pack aliphatic acrylic polyurethane	50
							Total	**260**
Steel Tanks inside Surface (Total) for Naphtha Storage	PS-4	Less than 50 deg C	SA 2.5	Primer	2	Shop	Zinc Rich Primer	40
				Finish	2	Site	2 Pack Epoxy Mastic	150

Surface/ Location	Paint Number	Temp	Surface prep	Coat	No. of coats	Application	Generic Type	NDFT/ Coat
							Total	**380**
Underneath of Bottom plate (Note 1)			SA 2	Primer	2	Shop	In Organic Zinc Silicate	40
				Finish	1	Site	FRP Phenolic Epoxy coating	125
							Total	205
Steel Tanks outside Surface (Total) for Naphtha		Less than 50 deg C	SA 2.5	Primer	1	Shop	Two pack inorganic zinc rich ethyl silicate /Two pack zinc rich epoxy paint	60
				Intermediate	1	Site	Two pack high build epoxy polyamide MIO	150
				Finish	1	Site	Two-pack aliphatic acrylic polyurethane	50
							Total	**260**
Steel Tanks inside Surface (Total) for Water Storage (Raw water, Potable and Distilled Water)	PS-5	Ambient	SA 2.5	Primer	1	Shop	Two component high build polyamide cured zinc phosphate Primer	75
				Finish	1	Site	Two component High build High Solid Modified Epoxy coating. (Suitable for potable water NSF 61 certified)	500
							Total	**575**
Underneath of Bottom plate		Ambient	SA 2	Primer	2	Shop	In Organic Zinc Silicate	40
				Finish	1	Site	FRP Phenolic Epoxy coating	125
							Total	205
Steel Tanks outside Surface (Total) for Water tanks		Ambient	SA 2.5	Primer	1	Shop	Two pack inorganic zinc rich ethyl silicate /Two pack zinc rich epoxy paint	60
				Intermediate	1	Site	Two pack high build epoxy polyamide MIO	150
				Finish	1	site	Two-pack aliphatic acrylic polyurethane	50
							Total	**260**
Un-Insulated piping and	PS-6	Less than	SA 2.5	Primer	1	*Site	Two pack inorganic zinc rich ethyl	60

Surface/ Location	Paint Number	Temp	Surface prep	Coat	No. of coats	Application	Generic Type	NDFT/ Coat
other components – Outside Surface (Including ACC duct)		100 deg C					silicate /Two pack zinc rich epoxy paint	
				Intermediate	1	*Site	Two pack high build epoxy polyamide MIO	150
				Finish	1	*Site	Two-pack aliphatic acrylic polyurethane	50
							Total	**260**
Insulated piping and other components – Outside Surfaces (Including condensate collection tank, flash tanks, blowdown tanks)	PS -7	Above 110 deg C	SSPC–SP6	Primer	1	*Site	Two pack inorganic zinc rich ethyl silicate	75
							Total	**75**
	PS -8	Below 110 deg C	SSPC–SP6	Primer	1	*Site	Two Pack Epoxy Primer	150
							Total	**150**
Pumps, Motors - Indoor	PS -9	Up to 120 Deg	SA 2.5	Primer	1	Shop	InOrganic Zinc Silicate	60
				Intermediate	1	Shop	High Build Epoxy	50
				Finish	1	Shop	Aliphatic Polyurethane	50
							Total	**160**
Pumps, Motors - Outdoor	PS -10	Up to 120 Deg	SA 2.5	Primer	1	Shop	Two pack inorganic zinc rich ethyl silicate /Two pack zinc rich epoxy paint	60
				Intermediate	1	Shop	Two pack high build epoxy polyamide MIO	150
				Finish	1	Shop	Two-pack aliphatic acrylic polyurethane	60
							Total	**270**
Bypass stack (External)	PS -11		SA 2.5	Primer	1	Shop	Solvent based inorganic Zinc primer	40
				Finish	1	Shop	Silicone acrylic	40
							Total	**80**

Note 1: If tank is provided with cathodic protection Paint shall be suitable to tank cathodic protection system.
Note 2: * marked item paint application at Site/Shop shall be decided based on Project Execution requirement.